SANJANA RANI

Aerospace Engineer | Mass Properties, CAD & Data Analysis | Python

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PROFESSIONAL SUMMARY
Aerospace engineer with a joint international MSc (France, Belgium, Germany), strong in CAD (SolidWorks, CATIA, Creo) and structured Python-based data analysis and reporting. Background spans CFD, experimental aerodynamics, and self-directed engineering-mechanics projects, with a consistent habit of verifying every result against an independent method before trusting it. Quick to pick up new tools and domains as a task requires.

Key Skills

CAD & Geometry Definition: SolidWorks, CATIA, Creo – Component Modeling
Programming & Data Analysis: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, MS Office (Excel), Notion (Certified: Advanced), Git/GitHub, Structured Reporting & Data Quality Checks
CFD & Simulation: OpenFOAM, ANSYS Fluent, SU2, XFOIL, GMSH, ParaView
Turbomachinery & Propulsion: Mean-Line Design, Blade Geometry (ParaBlade), Cascade CFD
Experimental Test Campaigns: Wind-Tunnel & Water-Channel Testing, Test-Rig Design & Assembly, Instrumentation, DAQ (DEWESoft), PIV
Languages: English (Fluent), French (Beginner), German (A1, improving)

Experience

Freelance CFD Engineer *Mar 2026 – Present*
CFD & Technical Task Development for AI Model Training
- Build and code well-defined technical problems with genuine reference solutions and correctness verifiers for AI model-training platforms, primarily in CFD/OpenFOAM and mechanical engineering domains.
- Deliver against reviewer-graded quality standards across multiple contracted platforms, iterating on structured feedback; numerous tasks accepted to date.

DLR Institute of Electrified Aero Engines (DLR e.V.), Cottbus, Germany *Feb 2024 - Sep 2024*
Graduate Research Assistant
- Conducted a literature review of electric propulsion systems for UAV/eVTOL applications to select an appropriately rated electric motor for the experimental test setup.
- Collaborated cross-functionally to configure and test a multi-sensor instrumentation array across four spatial positions using DEWESoft DAQ, assessing electric motor-induced EMI effects on microphones and accelerometers for next-generation LU(FT)^2 2030 eVTOL propulsion.
- Generated and processed a multi-microphone acoustic dataset (8 microphones across 6 varied experimental parameters) via a Python pipeline (FFT, coherence, magnitude spectra), analyzing coherence between acoustic and vibration (accelerometer) signals during motor operation – an aeroacoustics/vibroacoustics diagnostic approach – for master's thesis.

Chair of Aerodynamics and Fluid Mechanics, BTU, Cottbus, Germany *Aug 2023 – Feb 2024*
Student Assistant
- Conducted PIV and flow visualisation to characterize flow regimes (laminar, transitional, turbulent) across trial configurations; confirmed laminar flow achievement in final design and produced structured validation reports for department records.
- Designed and iteratively refined a cylindrical plate geometry in Creo – trial-and-error against target laminar-flow geometry constraints, across 3 configurations – then solely assembled the resulting experimental prototype rig (piping, optical/electrical components, laser/camera) for the TEKUS / DEPIK zero-gravity convection project.

Simulation Lab *May 2023 – Jul 2023*
Aerodynamics Intern (Freelance)
- Ran an 80-case ANSYS Fluent CFD sweep on a provided airfoil geometry (4 angles of attack $\times$ 20 deflection angles), computing lift/drag coefficients (C_L/C_D) to characterize AoA and deflection-angle effects on aerodynamic performance.

Education

Joint MSc in Transfer of Fluid Materials in Aeronautical & Space Applications (TFM-ASA) *Sep 2022 - Sep 2024*
Université de Bordeaux (Bordeaux, France) Université Catholique de Louvain (Louvain-la-Neuve, Belgium) Brandenburgische Technische Universität (Cottbus, Germany)
- **Thesis:** Electromagnetic Interference and its effects on Acoustic Sensor measurements
- **Selected Coursework:** Aerodynamics of External Flows, Gasdynamics & Reacting Flows, Introduction to Turbomachinery, Experiments in Aerodynamics & Fluid Mechanics, Flow Measurements

B.E. in Aerospace Engineering *Aug 2018 - Jun 2022*
Chandigarh University, Gharaun, India

- **Core Coursework:** Aerodynamics I & II, Aerothermodynamics, Propulsion, Wind Tunnel Testing, Flight Mechanics, Aircraft stability & control, Computational Fluid Dynamics
- **Final Project:** CFD Analysis of a hypersonic nozzle using ANSYS Fluent

Selected Projects

Composite Fuselage Laminate Analysis — **Self-directed** *Aug 2026*

- Built a Classical Lamination Theory toolkit: lamina/laminate stiffness matrices, ply-level stress/strain recovery, and failure assessment against three criteria (Maximum Stress, Tsai-Hill, Tsai-Wu) for fuselage-skin loading, with a design-space search finding a 25% mass reduction under a minimum-gauge constraint.
- Verified every mechanics stage against an independent derivation (166 automated tests) and explicitly labeled loads/strength data as illustrative, preliminary-design-level – not certified for production use.

Propellant Tank Wall Sizing — **Self-directed** *Aug 2026*

- Compared aluminum (Al 2219-T87), titanium (Ti-6Al-4V), and carbon/epoxy composite for a cylindrical propellant-tank wall with hemispherical domes under internal pressure: thin-wall membrane sizing for the metals, a from-scratch Classical Lamination Theory implementation (three failure criteria, bounded deterministic laminate search) for the composite.
- Found titanium and the composite effectively mass-equivalent (43.95 kg vs. 43.98 kg, 0.08% difference) while aluminum was 45% heavier under the same idealized model – verified by 324 automated tests.

Wing-Planform Design — **Self-directed** *Aug 2026*

- Designed two materially distinct trapezoidal wing-planform concepts (steady-cruise and passive-gust-relief) for a regional-aircraft mission, trading aspect ratio, sweep, taper, twist, dihedral, and torsional stiffness against rigid/flexible aerodynamic efficiency, stall, bending, and structural-mass constraints.
- Built the deterministic verification referee around a 9-term Prandtl lifting-line method, checking each concept's rigid and aeroelastic (flexible-wing) response and as-built-tolerance robustness across 256 corner-case geometry perturbations per concept.

Water-Channel PIV Study — Airfoil & Cylinder — BTU Cottbus *Aug 2023 – Feb 2024*

- Designed and iteratively refined a cylindrical plate geometry in Creo – trial-and-error against target laminar-flow constraints, across 3 configurations – then assembled it in CAD with existing piping, optical, and electrical component models (tolerances set by the supervising team, with some intuition-based input from me) and fabricated and physically assembled the resulting experimental prototype rig (laser/camera included).
- Conducted stereoPIV and flow visualization on the built hardware to confirm the design met its laminar-flow target, producing structured validation reports comparing as-built performance against design intent.

Extracurricular Activities

Aerodynamics Club, Chandigarh University – *2018 – 2022* – team-built RC aircraft and static wind-tunnel models as part of coursework projects

FlowSleuth — Agentic CFD Root-Cause Diagnostics, micro1 Frontier Engineering Challenge 2026 (Hackathon) – *Aug 2026* – built an agentic diagnostic tool for failed/stalled OpenFOAM CFD runs, improving root-cause accuracy from a 63.42/100 baseline to 98.48/100 across a 12-case benchmark